

Team Reflection

Project : 10

Project Title: Motor Modelling

Category: E

Team Members: Arshaan Lakhani(H036), Aadhar Patel(H050), Ishan Patel(H051), Paarth Patel(H051)

Date: 19/04/2026

Q1: What was the single hardest technical challenge you faced in deriving the mathematical model? How did you resolve it?

The project demanded time and efforts where we explored a lot of new things. First, we started by understanding the working of all the motors. We analyzed the electrical and mechanical working behind DC, Stepper and Servo motor which helped us understand the mathematical model and simulation. The most challenging part of our project was handling the difference in modeling approach in all 3 motors withing single framework. While the DC motors follow linear relationship between current, torque and speed. The stepper introduces non-linear due to its stepping and precise mechanism, while servo have feedback-based control and hence have different mechanical relation with respect to it's gears.

Q2: Which Course Outcome (CO1-CO5) was most challenging for your team? Why?

The most challenging outcome for our team was CO4: applying constitutive laws to analyse dynamic systems. CO4 was tough because it required us to effectively link electrical and mechanical relationships within each motor. We needed to use laws like torque-current relations and back EMF-speed dependence. Then, we had to integrate these into differential equations that represent system behaviour. This got more complicated when we dealt with different motor types, particularly the nonlinear features of the stepper motor and the feedback-driven nature of the servo motor.

Q3: What would you do differently if you started the project over?

There were many things which we realized while doing the project which could get better. Firstly we explored different motors and simulated it on the same system. However, the purpose of each motor is different and different simulation would help us relate better. We would also

manage to better explore real life implementation by studying existing simulation which would help us understand and implement the system in a better way.

Q4: What did you learn from this project that a regular textbook problem could NOT teach you?

This project gave us a practical knowledge of dynamic systems and models which we learn in classroom. The textbook problem defines the aspects of predefined solution while we approached open-ended engineering problem which helped us understand the impact of both electrical and mechanical system in a model. We also dealt with real world approximations and interpretation which is critical for understanding the implication of engineering in real world to be able to innovate and contribute in future.

Q5: How did you divide work among team members? Did everyone understand the full derivation?

We divided the work equally assigning each member a motor type to study and research. We then would use the respective classroom hours to exchange the information we gathered which would make everyone responsible for their work and also ignite collaborative learning and team work. All team members understood the complete modelling and simulation as the discussions often carried new ideas of implementation.

Q6: What AI tools did you use? For what specific purpose? What did you NOT use AI for?

We used AI tools such as

ChatGPT for:

- Clarifying theoretical concepts

Claude for:

- Debugging Python code
- Verifying intermediate steps in equations

Google Gemini:

- Pictorial representation of models and implementation of simulation

We did not rely on AI for FBD and final derivations. We deduced the forces by ourselves and used AI to correct us and see if we are on the right path.